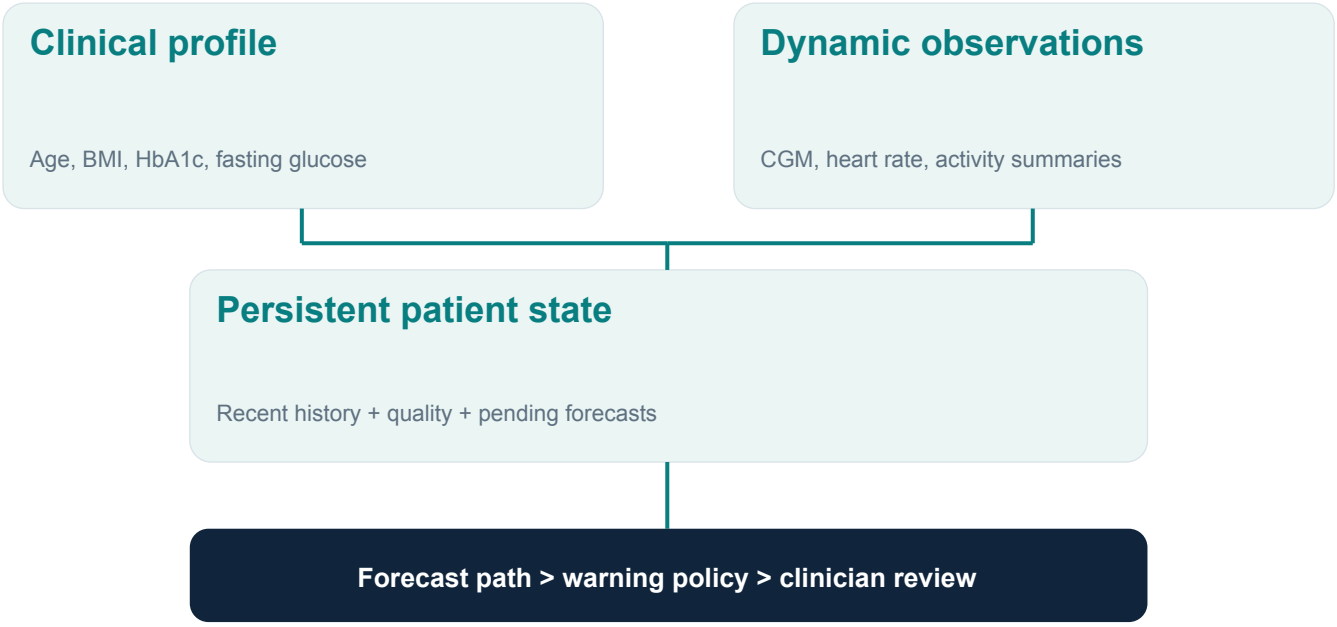
03 / ARCHITECTURE

A changing state with a forecast ledger



The patient state receives thirteen consecutive five-minute observations spanning an hour. Each update matures old forecast errors, checks the input history and emits estimates at +5 through +30 minutes. Missing required history produces a withheld state.

The evidence ledger

ForecastLedger records the model artifact hash, clinical-profile usage, received observations, issuance time and six target values. A SHA256 digest covers the fixed issued payload. Later outcomes append separately when actual target readings arrive. Missing targets remain unobserved and never become zero error.

Controlled delivery checks

The lab preserves the source timestamp of delayed packets. Ten-minute-old packets fail the five-minute freshness rule. The current input becomes unavailable, with no timestamp relabeling, carry-forward or retroactive backfill. Forecasts resume after thirteen fresh consecutive observations.

Clinical fusion engines use both clinical and dynamic fields. Validation selected the CGM-only engine and a bias-update rate of zero. The twin is a data-driven patient-state representation, with no validated physiological reconstruction or learned personalization benefit.